# Isaac Oliva-González

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## Summary

Graduate researcher in computer science with a chemical engineering background. My work focuses on integer and semidefinite programming, quantum algorithms, and symmetry reduction techniques for combinatorial optimisation. I develop and analyse mathematical programming models for large-scale problems, with applications in energy and process systems.

## Research Interests

**Optimization & Operations Research**

Combinatorial Optimization, Integer & Semidefinite Programming, Algebraic Reformulations, Graph-Theoretic Approaches

**Quantum Computing & Algorithms**

Hybrid Quantum-Classical Methods, Quantum Machine Learning, Quantum Annealing

**Applications**

Energy Systems, Process Systems Engineering, Large-Scale Optimisation

## Education

**MSc in Computer Science** *Aug 2023 - Aug 2025*

Universidad Autónoma de Querétaro Querétaro, MEX

- Dissertation on quantum annealing and simulated annealing for discrete-space route optimization.

**BSc in Chemical Engineering** *Jan 2017 - Dec 2021*

Universidad de Guanajuato Guanajuato, MEX

- Dissertation on modeling and design of intensified distillation sequences for renewable fuel separation.

## Experience

**Tecnológico de Monterrey** *Jan 2025 - Aug 2025*

Visiting Research Scholar (Remote) | *Supervisor: Dr. Ilse María Hernández-Romero* Monterrey, MEX

- Conducted collaborative research on hybrid quantum–classical neural architectures using variational quantum circuits to model black carbon dynamics, benchmarking against classical models to assess the contribution of quantum-enhanced architectures in predictive performance and model expressiveness (*PennyLane*). See [J1].

**Universidad Autónoma de Querétaro** *Aug 2023 - Aug 2025*

MSc Student Querétaro, MEX

- Designed and benchmarked quantum and classical optimisation algorithms for discrete shortest-path problems, analysing performance trade-offs across solution quality and computational efficiency (*Julia*). See [J2].
- Designed and implemented finite automata for encoding alphabetic sequences and performing operations on deterministic finite automata (*Mathematica*).

**Ingredion México** *Feb 2022 - Dec 2022*

Process Technician San Juan del Río, MEX

- Operated and monitored industrial starch processing systems, performing on-site troubleshooting and real-time adjustments to ensure process reliability.

**NOVARES México** *Jan 2021 - Jun 2021*

Undergraduate Intern Silao, MEX

- Supported the implementation of health, safety, and environmental (HSE) programs in an automotive manufacturing environment.

**Instituto Tecnológico de Villahermosa** *Summer 2019*

Visiting Research Scholar | *Supervisor: Dr. Francisco López-Villarreal* Villahermosa, MEX

- Developed a computational interface for vapor–liquid equilibrium calculations in isothermal flash distillation, applying numerical methods to thermodynamic optimization problems (*MATLAB*).

## Research Projects

**Symmetry Reduction in Combinatorial Optimization** *Julia, JuMP*
Available here 2026

- Exploratory study of symmetry reduction in combinatorial optimisation (ILP), including shortest path, graph coloring, max-cut, and TSP. Applied lexicographic constraints, variable fixing, and automorphism-based methods to reduce redundant solutions.

**Symmetry Reduction in Combined Cycle Unit Commitment** *Julia, JuMP*
Available here 2026

- Study of symmetry reduction applied to the *Combined Cycle Min-Up/Min-Down Unit Commitment Problem* (CC-MUCP). Derived demand-aware lexicographic constraints to break the two-level permutation symmetry from from identical packages and gas turbines.

## Skills

**Programming**: Python, Julia
**Optimisation & Modelling**: JuMP, GAMS, MATLAB
**Quantum & Scientific Computing**: PennyLane, Qiskit, Mathematica, Aspen Plus, Jupyter Notebooks

## Languages

**Spanish**: Native proficiency.
**English**: Proficient (TOEFL ITP 562 | CEFR B2).
**German**: Elementary proficiency (CEFR A2).

## Research & Publications

### Publications

[J1] **A Hybrid Quantum-Classical Machine Learning Framework for Black Carbon Forecasting** *EPJ Quantum Technol.*
**I. Oliva-González**, L. T. González, O. D. Lara-Montaño, A. I. Ramirez, A. Mendoza, I. M. Hernández-Romero 2026
*EPJ Quantum Technology* (*accepted*)

[J2] **QUBO-based simulated annealing approach for the shortest path problem with applications to urban transportation networks** *Algorithms*
**I. Oliva-González** and H. Jiménez-Hernández 2026
*Algorithms* (*under review*). Preprint available. DOI: 10.20944/preprints202511.0578.v1

### Book Chapters

[B1] **Integrating Quantum Computing into Sustainable Carbon-Capture Materials Research: Opportunities and Perspectives** *Hand. Carbon Neg. Footpr. Mater.*
I. M. Hernández-Romero, **I. Oliva-González**, O. D. Lara-Montaño, K. Hernández-Romero, G. G. Ezquivel-Patiño, O. Kharissova, A. I. Ramirez, L. T. González 2026
*Handbook of Carbon Negative Footprint Materials*, Springer Nature (*accepted*)

### Conference Proceedings

[C1] **Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación** *Av. Ing. Quím.*
**I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández 2024
*Avances en Ingeniería Química*, vol. 3, no. 1, pp. 265–270, AMIDIQ 2024. Available online

[C2] **Computer-aided design of intensified separation sequences for a complex mixture of renewable hydrocarbons** *Comput. Aided Chem. Eng.*
**I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández 2024
*ESCAPE34/PSE2024*, Elsevier, vol. 53, pp. 1483–1488, DOI: 10.1016/B978-0-443-28824-1.50248-9

[C3] **Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación** *Av. Ing. Quím.*
**I. Oliva-González**, C. Gutiérrez-Antonio, F. I. Gómez-Castro, S. Hernández, and A. G. Romero-Izquierdo 2023
*Avances en Ingeniería Química*, vol. 2, no. 1, pp. 402–407, AMIDIQ 2023. Available online

[C4] **Síntesis de secuencias intensificadas de destilación para la separación de una mezcla multicomponente de hidrocarburos renovables** *Av. Ing. Quím.*
**I. Oliva-González**, C. Gutiérrez-Antonio, F. I. Gómez-Castro, E. Quiroz Pérez, S. Hernández, and A. G. Romero-Izquierdo 2022
*Avances en Ingeniería Química*, vol. 1, no. 4, pp. 218–223, AMIDIQ 2022. Available online

## Conference Presentations

[P1] **Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación**
**I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández 2023
Retos y Oportunidades al Net-Zero 2030, Querétaro, México (Oral presentation)

[P2] **Synthesis of alternative intensified sequences for the separation of a multicomponent renewable hydrocarbons mixture**
**I. Oliva-González**, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández 2022
*12th International Conference Distillation & Absorption 2022* (DA2022), Toulouse, France (Poster presentation)